COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 2 Activity Worksheet | Instruction Formats, Instruction Flow, Addressing Modes & 8086
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1. Identifying Instruction Formats

• ADD R1, R2, R3 → Three-address format — the instruction names three operands (two sources R2, R3 and destination R1).

• MOV AX, BX → Two-address format — two operands are specified (source BX, destination AX).

• PUSH AX → One-address format — only one explicit operand (AX); the stack pointer is implied.

• ADD → Zero-address format — no explicit operands; operands are taken from the top of the stack (stack machine style).

• LOAD R1, #10 → Two-address format — a destination register (R1) and a source operand (immediate constant 10) are both specified.

2. Tracing Instruction Flow

• START: MOV AX, #5 → PC points to START; AX ← 5; PC advances to next instruction.

• ADD AX, #3 → AX ← 5 + 3 = 8; PC advances.

• JMP END_LOOP → PC is loaded with the address of END_LOOP (non-sequential jump).

• END_LOOP: HLT → processor halts.

• Note: the instructions at LOOP (INC AX; JMP START) are never executed, because control jumps directly from ADD to END_LOOP; the PC therefore follows the sequence START → (next) → END_LOOP instead of incrementing through every line.

3. Addressing Mode Scenarios

• Loading a constant value directly into a register → Immediate addressing — the operand is part of the instruction, so no memory access is needed and it is fast.

• Accessing an array element using an index stored in another register → Indexed addressing — effective address = base address + index register, ideal for stepping through arrays.

• Retrieving data from a memory location whose address is stored in a register → Register indirect addressing — the register holds the pointer, allowing flexible access to computed locations.

• Jumping to a subroutine whose address is known at assembly time → Direct (absolute) addressing — the target address is encoded directly in the instruction.

• Accessing a variable located at a fixed memory address → Direct addressing — the instruction contains the fixed address of the variable, requiring only one memory reference.

4. 8086 Architecture Diagram (described)

• The 8086 is split into two units. Bus Interface Unit (BIU): contains the segment registers (CS, DS, SS, ES), Instruction Pointer (IP), address-generation logic (physical address = segment × 16 + offset), the 6-byte instruction queue and the bus control logic — it fetches instructions and data from memory.

• Execution Unit (EU): contains the general-purpose registers (AX, BX, CX, DX, SP, BP, SI, DI), the ALU, the FLAG register and control circuitry — it decodes and executes the queued instructions.

• The two units work in parallel (pipelining): while the EU executes, the BIU prefetches the next instructions into the queue.